

06 - Splunk Private

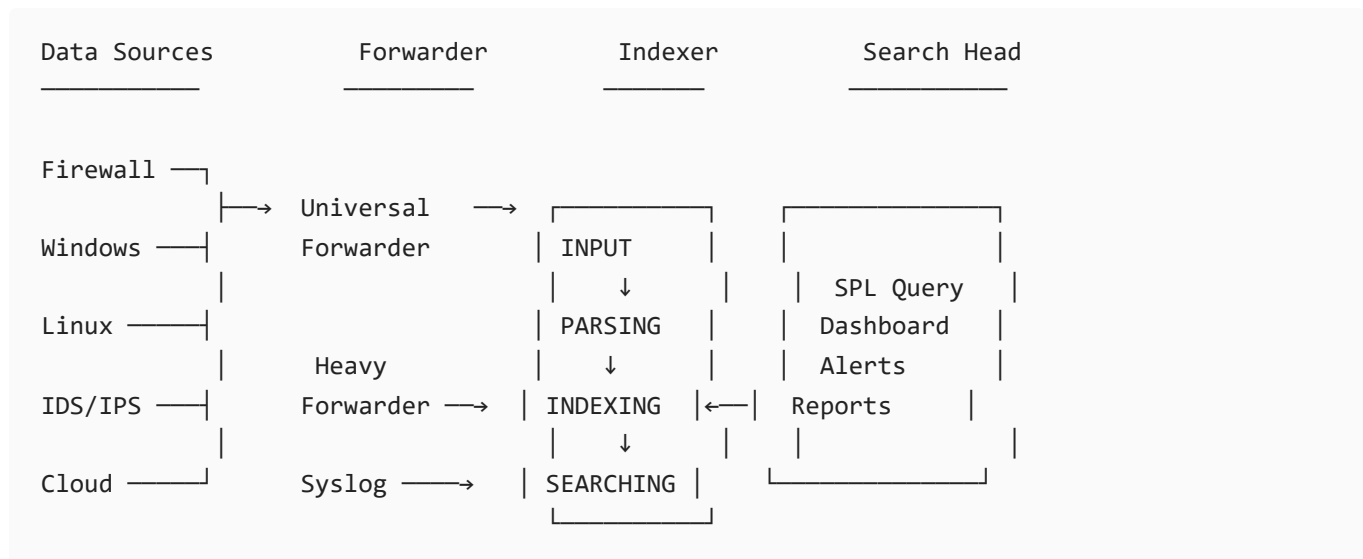
Splunk Enterprise is a SIEM & log analysis platform used to:

- Collect logs
- Search logs
- Analyze events
- Create dashboards
- Detect threats
- Generate alerts

SOC analysts use Splunk daily to:

- Investigate alerts
- Hunt threats
- Monitor suspicious activity
- Create detection rules

Endpoint → Forwarder → Indexer → Search Head → SOC Analyst



Splunk Architecture

Core Components:

1) Forwarder

- Installed on endpoints (Windows/Linux servers)
- Collects logs
- Sends logs to Indexer

Types:

- Universal Forwarder (lightweight)
- Heavy Forwarder (can parse/filter logs)

2) Indexer

- Receives logs
- Parses logs
- Stores logs in indexes
- Makes data searchable

Think of it as:

Storage + Processing Engine

3) Search Head

- Where users log in
 - Run searches (SPL)
 - Create dashboards
 - Create alerts
-

Index

An Index is where logs are stored.

Examples:

- index=windows
- index=firewall
- index=proxy
- index=linux

SOC analysts always start search with:

```
index=windows
```

Splunk Data Structure

Every event has:

- `_time` → timestamp
- `host` → system name
- `source` → log file path
- `sourcetype` → log type
- `index` → storage location

Example:

```
index=windows sourcetype=WinEventLog:Security
```

SPL (Search Processing Language)

SPL is how you search logs.

Basic Search

```
index=windows
```

Search all Windows logs.

Keyword Search

```
index=windows "failed"
```

Field Based Search

```
index=windows EventCode=4625
```

4625 = Failed login

stats

Used to aggregate data.

Example:

```
index=windows EventCode=4625  
| stats count by user
```

Shows failed login count per user.

top

```
index=windows EventCode=4625  
| top user
```

table

```
index=windows EventCode=4688  
| table _time host user process_name
```

sort

```
index=windows EventCode=4688  
| sort - count
```

where

```
index=windows EventCode=4688  
| where count > 10
```

Creating Alerts

Example: Alert for brute force

```
index=windows EventCode=4625  
| stats count by user  
| where count > 20
```

Save As → Alert

Trigger → If results > 0

Dashboards in Splunk

Dashboards are visual panels (charts, tables, graphs).

Used in SOC to monitor:

- Failed logins
- Top talkers

- Malware alerts
- Firewall denies

How to Create Dashboard

1. Run query
2. Click Visualization
3. Choose chart type (bar, pie, line)
4. Save As → Dashboard panel

Example:

```
index=windows EventCode=4625  
| stats count by host
```

Visualize as bar chart.

Splunk Apps

1) Splunk Enterprise Security

- SIEM layer
 - Correlation searches
 - Risk scoring
 - Incident review
- Used in enterprise SOC.

2) Splunk SOAR

- Automation
- Playbooks
- Auto response